# GIULIA PRETI

## CONTACT INFO


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## ABOUT ME

I am a **Data Scientist** with a PhD in Computer Science and over five years of experience applying machine learning and graph mining techniques to academic research and industry projects. My expertise lies in identifying significant and anomalous structures in complex data, and developing null models for attributed networks, with applications in portfolio optimization and anti–money laundering. Experienced in academic and industry collaborations, I excel at managing multiple projects and translating complex findings into actionable insights for diverse stakeholders. I am eager to continue advancing research in graph-based learning while partnering with companies to address challenging real-world problems.

## EXPERIENCE

### SCIENTIFIC RESEARCHER — 2022 -

***CENTAI*** | *Turin, Italy*

- EU Project: Common European Data Spaces and Robust AI for Transparent Public Governance (cedar-heu-project.eu).
- Industrial Project with Isybank: Predictive Understanding for Loyalty, Segmentation, and Engagement.
- Industrial Project with Isybank: Policy Optimization for Customer Outreach and Recommendation Notifications.
- Industrial Project with ISP and Enel: Correlation-based Hedging Strategies for Illiquid Commodity Futures and Market-neutral Risk Mitigation Strategy for Long-term Illiquid Contracts.
- Research Projects: Null Models for Hypergraphs and Labeled Graphs, Graph Anomaly Detection, Significant Pattern Mining, Data Summarization.

### POSTDOCTORAL RESEARCHER — 2020 - 2022

***ISI Foundation*** | *Turin, Italy*

- Research Projects: Data mining for graphs and higher-order networks.
- Industrial Project with ISP: Portfolio optimization and volatility forecasting.

### TEACHING ASSISTANT — 2015 - 2018

***University of Trento*** | *Trento, Italy*
(Computability and Computational Complexity course)

**VISITING RESEARCHER** **May 2018 - Sep 2018**

***Aalto University School of Science and Technology*** | *Espoo, Finland*

- Research Topic: event detection in dynamic networks.

**VISITING RESEARCHER** **Jul 2017 - Sep 2017**

***Laboratoire de Recherche en Informatique (LRI)*** | *Orsay, France*

- Research Topic: entity matching in heterogeneous datasets.

## EDUCATION

**PhD in Computer Science** **2015 - 2019**

***University of Trento*** | *Trento, Italy*

**MSc in Computer Science** **2013 - 2015**

***University of Trento*** | *Trento, Italy*
(Graduated *Summa Cum Laude*)

**BSc In Mathematics** **2009 - 2013**

***University of Trento*** | *Trento, Italy*
(Graduated 106/110)

## ACHIEVEMENTS

**Awards**
- Best reviewer at ACM KDD 2024 and The Web Conference 2021.

**Scientific Community**
- Invited speaker at WoRSCH (2025) and Dagstuhl Seminar 24391 (2024).
- Appointed Area Chair for KDD (2025, 2026).
- Appointed Tutorial Chair for CIKM 2025.
- Appointed PhD Consortium Chair for KDD (2024, 2026).

## SKILLS

**Areas of Expertise**
- Machine Learning, Graph Mining, and Network Analysis.
- Financial Data Mining and Anomaly Detection.

**Technical Skills**
- Advanced: Python, Java, Git, LaTeX, Microsoft Office.
- Familiar: Bash, Spark, Scala, Web technologies.

**Soft Skills**
- **Project Management and Prioritization**: managing multiple projects simultaneously.
- **Cross-Functional Collaboration**: organizing tasks in a team and working in multicultural environments.
- **Data Translation and Presentation**: translating complex findings into actionable insights and presenting research results.

**Languages**
- Italian (native)
- English (Full professional proficiency)

## PUBLICATIONS

### Conference Papers

Abrate, Carlo, Giulia Preti, and Francesco Bonchi (2023). "Counterfactual explanations for graph classification through the lenses of density". In: *World Conference on Explainable Artificial Intelligence*. Springer, pp. 324–348.

Preti, Giulia, Gianmarco De Francisci Morales, and Francesco Bonchi (2021). "Strud: Truss decomposition of simplicial complexes". In: *The Web Conference*, pp. 3408–3418.

Preti, Giulia, Gianmarco De Francisci Morales, and Francesco Bonchi (2022). "Fresco: mining frequent patterns in simplicial complexes". In: *ACM Web Conference*, pp. 1444–1454.

Preti, Giulia, Gianmarco De Francisci Morales, and Matteo Riondato (2021). "MaNIACS: Approximate Mining of Frequent Subgraph Patterns through Sampling". In: *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 1348–1358.

Preti, Giulia, Matteo Lissandrini, Davide Mottin, and Yannis Velegrakis (2018). "Beyond Frequencies: Graph Pattern Mining in Multi-weighted Graphs". In: *International Conference on Extending Database Technology*, pp. 169–180.

Preti, Giulia, Matteo Riondato, Aristides Gionis, and Gianmarco De Francisci Morales (2025). "Polaris: Sampling from the Multigraph Configuration Model with Prescribed Color Assortativity". In: *ACM International Conference on Web Search and Data Mining*, pp. 30–39.

Preti, Giulia, Polina Rozenshtein, Aristides Gionis, and Yannis Velegrakis (2021). "Discovering Dense Correlated Subgraphs in Dynamic Networks". In: *Pacific-Asia Conference on Knowledge Discovery and Data Mining*. Springer, pp. 395–407.

Rozenshtein, Polina, Giulia Preti, Aristides Gionis, and Yannis Velegrakis (2020). "Mining dense subgraphs with similar edges". In: *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, pp. 20–36.

### Journal Articles

Preti, Giulia, Gianmarco De Francisci Morales, and Francesco Bonchi (2024). "Hyper-distance oracles in hypergraphs". In: *The VLDB Journal* 33.5, pp. 1333–1356.

Preti, Giulia, Gianmarco De Francisci Morales, and Matteo Riondato (2023). "Maniacs: Approximate mining of frequent subgraph patterns through sampling". In: *ACM Transactions on Intelligent Systems and Technology* 14.3, pp. 1–29.

Preti, Giulia, Gianmarco De Francisci Morales, and Matteo Riondato (2024). "Impossibility result for Markov chain Monte Carlo sampling from microcanonical bipartite graph ensembles". In: *Physical Review E* 109.5, p. L053301.

Preti, Giulia, Adriano Fazzone, Giovanni Petri, and Gianmarco De Francisci Morales (2024). "Higher-order null models as a lens for social systems". In: *Physical Review X* 14.3, p. 031032.

Preti, Giulia, Gianmarco de Francisci Morales, and Matteo Riondato (2024). "Alice and the Caterpillar: A more descriptive null model for assessing data mining results". In: *Knowledge and Information Systems* 66.3, pp. 1917–1954.

Preti, Giulia, Matteo Lissandrini, Davide Mottin, and Yannis Velegrakis (2021). "Mining patterns in graphs with multiple weights". In: *Distributed and Parallel Databases* 39.2, pp. 281–319.

**Demos**

Preti, Giulia, Polina Rozenshtein, Aristides Gionis, and Yannis Velegrakis (2019). "Excode: a tool for discovering and visualizing regions of correlation in dynamic networks". In: *International Conference on Data Mining Workshops*. IEEE, pp. 1114–1117.